

SyscallAssignment – 31567

```
sufyan_sohail@sufyan-HP:/mnt/c/Users/sufya/Documents/GitHub/xv6-labs-2026$ cd /mnt/c/Users/sufya/Documents/GitHub/xv6-labs-2026
sufyan_sohail@sufyan-HP:/mnt/c/Users/sufya/Documents/GitHub/xv6-labs-2026$ gdb-multiarch
GNU gdb (Ubuntu 15.1-1ubuntu1~24.04.1) 15.1
Copyright (C) 2024 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
Type "show copying" and "show warranty" for details.
This GDB was configured as "x86_64-linux-gnu".
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
<http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word".
The target architecture is set to "riscv:rv64".
warning: No executable has been specified and target does not support
determining executable automatically. Try using the "file" command.
0x0000000000010000 in ?? ()
(gdb) b syscall
Breakpoint 1 at 0x000001d52: file kernel/syscall.c, line 138.
(gdb) c
Continuing.
[Switching to Thread 1.3]

Thread 3 hit Breakpoint 1, syscall () at kernel/syscall.c:138
138 {
(gdb) backtrace
#0  syscall () at kernel/syscall.c:138
#1  0x0000000000001b0c in usertrap () at kernel/trap.c:68
#2  0x00000003ffffff09c in ?? ()
(gdb) |
```

Lab Q1) The **usertrap()** function in **kernel/trap.c** calls the **syscall()**

```
(gdb) n
140 struct proc *p = myproc();
(gdb) n
142 num = p->trapframe->a7;
(gdb) p /x p->trapframe->a7
$1 = 0xf
(gdb) p p->name
$2 = "init", '\000' <repeats 11 times>
(gdb) p p->pid
$3 = 1
(gdb) |
```

```
sufyan_sohail@sufyan-HP:/mnt/c/Users/sufya/Documents/GitHub/xv6-labs-2026$ grep -n "SYS_open\|SYS_exec" kernel/syscall.h
8:#define SYS_exec 7
16:#define SYS_open 15
sufyan_sohail@sufyan-HP:/mnt/c/Users/sufya/Documents/GitHub/xv6-labs-2026$ |
```

Q2) Q2: p->trapframe->a7 = 0xf (15), which is SYS_open. This is the syscall number for open(). The first user program xv6 starts is init (pid 1).

```
(gdb) p /x $sstatus
$4 = 0x200000022
(gdb) p /x ($sstatus >> 8) & 1
$5 = 0x0
(gdb) |
```

Q3) The previous mode was User mode. This is confirmed by the SPP bit (bit 8) of sstatus being 0, which means the CPU was in user mode before the trap into the kernel. The full sstatus value was 0x200000022.

```
xv6 kernel is booting
hart 2 starting
hart 1 starting
scause=0xd sepc=0x8001d62 stval=0x0
panic: kerneltrap
QEMU: Terminated
sufyan_sohail@sufyan-HP:/mnt/c/Users/sufya/Documents/GitHub/xv6-labs-2026$ grep "8001d62" kernel/kernel.asm
8001d62: 00002683 lw a3,0(zero) # 0 <_entry-0x80000000>
sufyan_sohail@sufyan-HP:/mnt/c/Users/sufya/Documents/GitHub/xv6-labs-2026$
```

Q4) Q4: The faulting instruction is "lw a3,0(zero)"

The register a3 corresponds to the variable num in the C code. The kernel crashes because address 0 is not mapped in the kernel address space. scause=0xd is a load page fault. stval=0x0 confirms the faulting address was 0.

```
(gdb) b *0x8001d62
Breakpoint 1 at 0x8001d62: file kernel/syscall.c, line 142.
(gdb) c
Continuing.
[Switching to Thread 1.3]
Thread 3 hit Breakpoint 1, syscall () at kernel/syscall.c:142
142 num = *(int*)0;
(gdb) p p->name
$1 = "init", '\000' <repeats 11 times>
(gdb) p p->pid
$2 = 1
(gdb) |
```

Q5: The process running when the kernel panicked was "init" with pid 1.

This is because with the injected bug (num = *(int*)0), the very first syscall made by any process dereferences address 0 and panics. init is the first user process xv6 starts, and its first syscall triggers the fault.

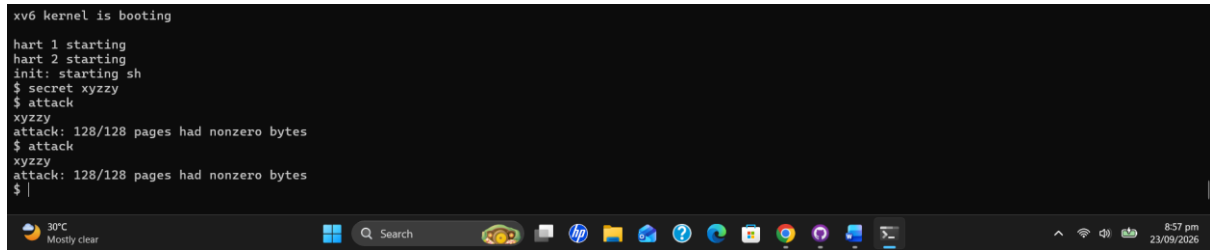
```
xv6 kernel is booting
hart 2 starting
hart 1 starting
init: starting sh
$ sandbox 32768 - cat README
cat: cannot open README
$ |
```

Exercise 2 Working

```
xv6 kernel is booting
hart 1 starting
hart 2 starting
init: starting sh
$ sandbox 32768 README grep xv6 README
xv6 is a re-implementation of Dennis Ritchie's and Ken Thompson's Unix
Version 6 (v6). xv6 loosely follows the structure and style of v6,
xv6 is inspired by John Lions's Commentary on UNIX 6th Edition (Peer
(Kaashoek,rtm@mit.edu)). The main purpose of xv6 is as a teaching
$ sandbox 32768 README grep xv6 x
grep: cannot open x
$ |
```

Exercise 3 Working

```
xv6 kernel is booting
hart 1 starting
hart 2 starting
init: starting sh
$ secret xyzzy
$ attack
xyzzy
attack: 128/128 pages had nonzero bytes
$ attack
xyzzy
attack: 128/128 pages had nonzero bytes
$ |
```

A screenshot of a Windows desktop environment. A terminal window is open, displaying the output of an xv6 kernel boot. The text in the terminal shows the kernel booting, two harts starting, and an initial shell prompt. A secret 'xyzzy' is entered, followed by an 'attack' command. The output indicates that 128 out of 128 pages had nonzero bytes. The Windows taskbar is visible at the bottom, showing the Start button, a search bar, and several application icons including File Explorer, Edge, and Chrome. The system tray on the right shows the date and time as 8:57 pm on 23/09/2026.

Exercise 4 Working